Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B.Tech | *Time: 03 Hours*

Code: 102603 | *Subject: Manufacturing Technology* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.

(ii) There are NINE questions in this paper.

(iii) Attempt FIVE questions in all.

(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 × 7 = 14]**

(a) Which of the following is NOT a non-conventional machining process?
(i) Electrical Discharge Machining (EDM)
(ii) Laser Beam Machining (LBM)
(iii) Milling
(iv) Electrochemical Machining (ECM)

(b) In mould and die design, which material is most commonly used for high-volume production due to its wear resistance?
(i) Mild Steel
(ii) High-Speed Steel
(iii) Tool Steel
(iv) Aluminium Alloy

(c) Which one of the following is a function unique to a jig, but not a fixture?
(i) Holding the workpiece.
(ii) Locating the workpiece.
(iii) Guiding the cutting tool.
(iv) Facilitating the use of less skilled labor.

(d) The space per side between the punch and die cutting edges in a press tool is called:
(i) Shear angle.
(ii) Clearance.
(iii) Die land
(iv) Angular relief.

(e) Which instrument is most commonly used for precise angular measurement?
(i) Vernier caliper
(ii) Sine bar
(iii) Micrometer
(iv) Dial gauge

(f) Which metrology instrument is typically preferred for non-contact, high-resolution 3D mapping of the complete surface topography (surface integrity) of a finely machined component?
(i) Stylus Profilometer (Contact)
(ii) Coordinate Measuring Machine (CMM)
(iii) Scanning White Light Interferometer (SWLI)
(iv) Optical Comparator

(g) In interferometry, the principle used for measurement is based on:
(i) Reflection of light
(ii) Refraction of light
(iii) Interference of light waves
(iv) Diffraction of light

(h) Which parameter is most critical for assessing surface integrity after machining?
(i) Hardness
(ii) Surface roughness
(iii) Grain size
(iv) Cutting speed

(i) In microscale machining, process metrology primarily focuses on:
(i) Dimensional accuracy at macro level
(ii) Tool life prediction only
(iii) Nano-level surface finish and micro-feature accuracy
(iv) Reducing cutting fluid usage

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(j) In Electrical Discharge Machining (EDM), material removal occurs due to:

(i) Mechanical abrasion

(ii) Electrochemical dissolution

(iii) Thermal erosion caused by electrical sparks

(iv) Laser heating

Q.2 (a) Explain the differences between conventional and non-conventional machining processes with suitable examples. [7]

(b) Describe the key considerations in mould and die design for plastic injection molding. [7]

Q.3 (a) Explain the core principle of 3-2-1 location used in the design of jigs and fixtures. Illustrate with a simple sketch how the six degrees of freedom are restricted for a rectangular block. [7]

(b) Explain the difference between blanking and piercing operations regarding the component size and the application of cutting clearance. State the general rule for applying the cutting clearance to the die and the punch for each operation. [7]

Q.4 (a) Define rake angle and clearance (or relief) angle for a single-point cutting tool. Briefly explain the function of each angle. [7]

(b) Define flash and gutter in the context of impression forging die design. Explain the purpose of the flash and gutter. [7]

Q.5 (a) Explain the concept of limits, fits, and tolerances with suitable examples. Why are they important in manufacturing? [7]

(b) What is interferometry in metrology? Explain its application in surface measurement and flatness checking. [7]

Q.6 (a) What is tolerance analysis in manufacturing and assembly? Why is it important for high-precision components? [7]

(b) Describe the testing methods used for alignment and surface integrity in precision machining. [7]

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Q.7 (a) Explain the principle and process parameters of Abrasive Jet Machining (AJM). What are its advantages and limitations? [7]

(b) Compare Water Jet Machining (WJM) and Abrasive Water Jet Machining (AWJM) in terms of working principle, applications, and limitations. [7]

Q.8 (a) Describe the principle of Ultrasonic Machining (USM) and discuss the effect of process parameters on Material Removal Rate (MRR). [7]

(b) Explain the working principle of Electrical Discharge Machining (EDM). Discuss the role of dielectric fluid and power circuits. [7]

Q.9 (a) Discuss Electrochemical Machining (ECM) with reference to etchant, maskant, process parameters, and applications. [7]

(b) What is selective assembly? Explain the concept, procedure and application of selective assembly in precision manufacturing industries. [7]

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